



OPEN Evaluating the utility of data integration with synthetic data and statistical matching

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Data integration enhances dataset utility but raises privacy concerns due to increased disclosure risks. Synthetic data offers a potential solution, though its role in data integration has not been thoroughly investigated. This study assesses synthetic data integration by evaluating the impact of varying common variables during statistical matching and exploring synthetic-real dataset combinations in donor-recipient settings. We used data from the Korean Genome and Epidemiology Study (KoGES) cohort, with the full dataset as the donor and one-quarter of the subjects as the recipient. Multiple synthetic datasets were generated from both datasets, with varying sets of common variables. Statistical matching was conducted using the nearest-neighbor hotdeck method. Data utility was evaluated using confidence interval overlap measures in the hazard ratio estimates under clinical scenarios to predict diabetes onset. When both donor and recipient data were synthetic, the all-available matched data generally outperformed other matching conditions. However, clinically relevant matching variables occasionally showed equivalent performances. The synthetic data showed comparable model accuracy to real data, although further investigation is warranted to understand the performance differences. Statistically matched synthetic data offers utility comparable to real data, providing a potential approach for reducing privacy risks while maintaining data utility.

Keywords Synthetic data, Biomedical data, Statistical matching

The demand for integrating diverse biomedical data sources is growing as it enables the generation of more comprehensive information¹. Two main methods are used in the data integration process. Record linkage using a (quasi-)identifier aims to identify records from the same units; however, it can increase the likelihood of privacy leakage, often requiring legal and administrative constraints. Conversely, statistical matching pairs are similar but usually unidentical. Although statistical matching involves lower privacy leakage, it may produce a less accurate estimate than record linkage using an identifier².

We focus on synthetic data to enhance data utility while protecting data privacy. Synthetic data mimics real data's structure and distribution³. The concept of generating synthetic data using multiple imputations was initially introduced to protect administrative data^{4,5}. Synthetic data, first applied to survey data, have rapidly emerged as a privacy protection method in the biomedical research field⁶. A notable application in healthcare involves the construction of realistic synthetic patient data that exhibit similar distribution properties and follow the standard data format⁷. For example, access to large cohorts, such as the N3C cohort from the USA and the UK Biobank from England, may be restricted due to privacy and security concerns. To address such challenges, synthetic data has been developed⁸. Consequently, the methods for generating synthetic data have diversified^{9,10}.

Despite the growing interest in synthetic data for privacy protection, there is limited research on its utility in data integration, particularly when using statistical matching methods. This study aims to assess the feasibility of combining synthetic data with statistical matching to maximize data utility in realistic clinical scenarios. The main goal is to evaluate whether statistical matching on synthetic data yields results comparable to real-world datasets. With the increasing prevalence of diabetes imposing a significant clinical and economic burden on

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the healthcare system, it is crucial to identify the relevant risk factors for the onset of the disease. Thus, our study links longitudinal diabetes claims data with cross-sectional survey data¹¹. The generation of synthetic versions of the outcome-linked survey data offers significant advantages, such as broader access to the dataset while preserving confidentiality. We hypothesize that integrating synthetic data through appropriate statistical matching could provide a promising approach for harnessing the full potential of data while addressing privacy concerns and data generation challenges.

Methods

Real data construction

We used longitudinal data from the Korean Genome and Epidemiology Study (KoGES), a large cohort study primarily designed to identify gene-environment interactions and their effects on common chronic diseases¹². With the baseline recruitment of 10,030 Korean adults older than 40 in 2001–2002 and biennial follow-up data up to 2017–2018 collected through health checkups until 2017–2018, the KoGES represented 18 years of longitudinal data. We used the median split homeostasis model assessment of insulin resistance (HOMA-IR) and median split beta cell function (HOMA-B) values from the original data to define insulin resistance and secretion cutoffs, respectively¹³. Body mass index (BMI) scores below 25 and waist circumferences (WCs) less than 90 cm in men or 85 cm in women were used as cutoffs¹⁴. In the baseline examination, we included subjects with no missing values for the variables of interest and excluded subjects with self-reported diabetes or high-risk subjects by applying glucose < 126 mg/dL and insulin levels > 1 μ U/mL. Personal identification information was excluded from this study, and the Institutional Review Board waived the requirement for informed consent. This study was approved by the Institutional Review Board of Seoul National University Bundang Hospital (IRB no. X-2204-748-903) and was conducted in accordance with relevant guidelines and regulations, including the Declaration of Helsinki.

Research scenarios

We selected two research scenarios to define the risk factors of diabetes onset. The first scenario focused on insulin resistance and impaired insulin secretion^{15,16}. Type 2 diabetes development is often linked to high insulin resistance and impaired insulin secretion, measured by a high HOMA-IR and a low HOMA-B, both calculated based from fasting insulin and glucose levels. The second scenario examined the combined effects of BMI and WC^{17,18}. Elevated BMI and waist circumference are well-established risk factors for the development of diabetes.

Synthetic data generation

The *synthpop* R package was used to generate synthetic data through a classification and regression trees (CART) method, which synthesizes each variable sequentially by sampling from a conditionally structured tree, ensuring that the variables are generated one at a time based on previously synthesized values¹⁹. This nonparametric method does not require distribution assumptions. For example, to generate synthetic diabetes onset events and time-to-event data, *synthpop* fitted a tree-structured survival model using binary recursive partitioning and drew samples from the values at the tree nodes. The example of the *synthpop* code is illustrated in the Supplementary document. Synthetically generated covariates included age, sex, smoking status, height, weight, hypertension, glucose, hemoglobin, total cholesterol (hereafter referred to as cholesterol), insulin, HbA1c levels, diabetes onset, and time to diabetes onset. Synthetic datasets were generated ten times using from the entire dataset and from $R_{25\%}$, referred to as $S_{100\%}^n$ and $S_{25\%}^n$, respectively (Fig. 1). Since $S_{25\%}^n$ was derived from $R_{25\%}$, it preserves the statistical structure of the real dataset while still having new synthetic values. In other words, due to the nature of the synthesizer, $S_{100\%}^n$ and $S_{25\%}^n$ exhibit similar conditional multivariate structures but contain distinct values. Therefore, unless otherwise specified, a randomly selected quarter of the data ($R_{25\%}$) was regarded as real data.

Statistical matching method

In statistical matching, common variables in the donor and recipient datasets are used to link corresponding records and generate a matched dataset. Conventionally, a larger dataset is designated as the donor, sharing a subset of its variables with a smaller recipient. In our experiments, the donor and recipient datasets both contain the same set of variables, making the matching and replaced features structurally identical²⁰. We conducted statistical matching with the full dataset as the donor and one-quarter of the subjects as the recipient. We chose a quarter split to balance the sizes of the recipient and donor datasets. A smaller recipient dataset could limit the sample size and restrict the generalizability of the results, while a larger recipient dataset would reduce the data available for the donor, making statistical matching less effective. Five matching schemes were compared by varying the combinations of real and synthetic data as donor-recipient datasets (Fig. 1). For example, $R_{75\%} \rightarrow R_{25\%}$ indicated that three-quarters of the real dataset served as the donor and were matched to a quarter of the real data as the recipient. Similarly, $R_{75\%} \rightarrow S_{25\%}^n$ denoted that three-quarters of the real dataset were used as the donor, while for a quarter of the real dataset, synthetic datasets were used as the recipients. For multiple synthetic data, matching between a real and synthetic dataset was conducted ten times. Additionally, $R_{100\%} \rightarrow R_{25\%}$ was used as a reference for exact matching, in order to illustrate how the selection of more variables affected the tightness of matching. We applied nearest-neighbor one-to-one optimal matching, which is a popular statistical matching method²¹. The Gower distance, which accounts for both numerical and categorical variables, was calculated to measure the similarity between records in the donor and recipient sets. The closest pairs were selected to minimize the total distance between matched records. The *StatMatch* R package was used to implement the statistical matching scheme²².

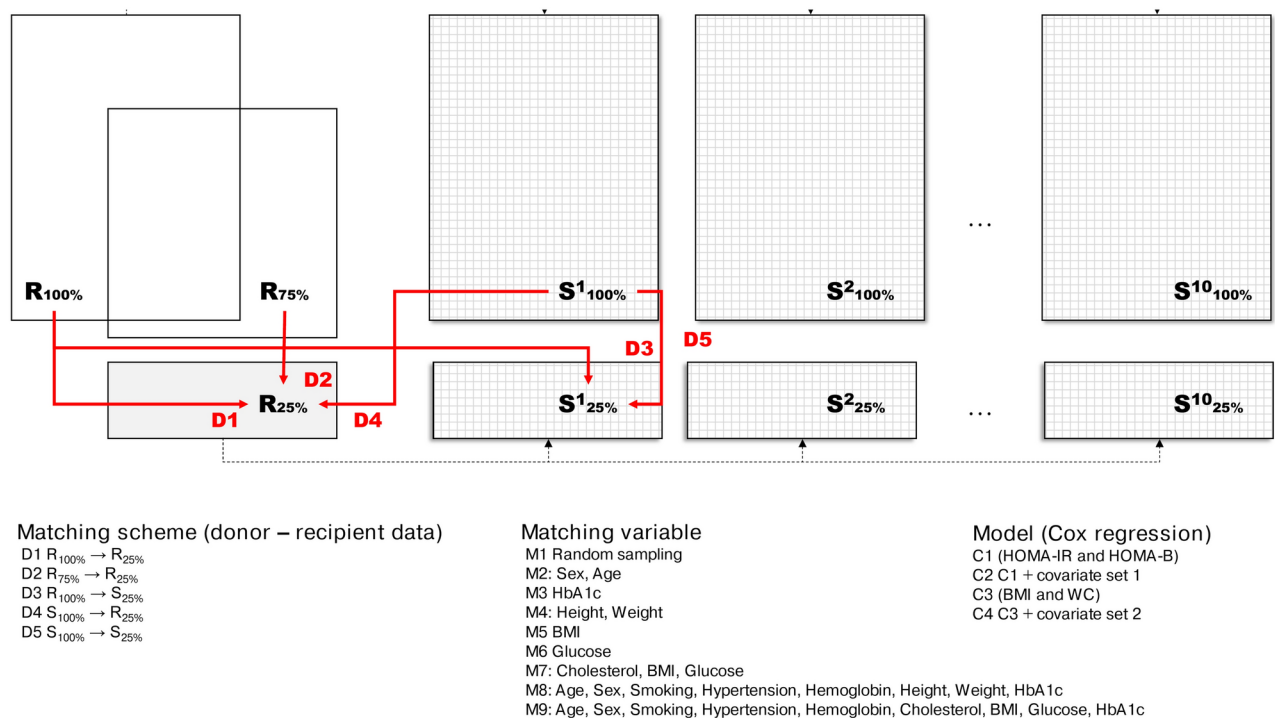


Fig. 1. Study schema. Synthetic datasets were generated ten times for the entire dataset and a randomly selected quarter of the data. Various matching schemes and variables were used. Two research scenarios were considered preceding the onset of diabetes: insulin resistance and impaired insulin as measured using homeostasis model assessment of insulin resistance (HOMA-IR and HOMA-B) and the combination of BMI and WC.

Matching variables setting

It is well-established that the quality of statistical matching largely depends on the selection of appropriate matching variables. To explore the effect of varying combinations of matching variables on data utility, we selected clinically important variables that are strongly associated with the onset of diabetes. The variables used in statistical matching were age, sex, smoking, hypertension, glucose, hemoglobin, cholesterol, height, weight, and HbA1c. To determine the optimal selection of matching variables, we tested nine different combinations: M1, random matching; M2, sex and age; M3, HbA1c; M4, height and weight; M5, BMI; M6, glucose; M7, cholesterol, glucose, and BMI; M8, age, sex, smoking, hypertension, hemoglobin, cholesterol, height, weight, and HbA1c; and M9, age, sex, smoking, hypertension, hemoglobin, cholesterol, BMI, glucose, and HbA1c. We expected that random matching for M1 would result in poor performance, while for M2, matching age and sex would reflect a common real-world approach. M3–M7 represented the matching schemes with a few clinically relevant variables for diabetes. M8 and M9 were designed to include a broader set of matching variables.

Coherence of distributions

We evaluated the fidelity of the synthetic data compared to real data and compared the variables of the matched records in the donor and recipient datasets. For instance, when comparing the synthetic and real data, the distribution of height in the real dataset $R_{25\%}$ was compared with that in each of the ten synthetic datasets $S^1_{25\%}$ to $S^{10}_{25\%}$. Similarly, when comparing donor and recipient data, the height distribution in the donor data was compared with that in the recipient data. In our simulation scheme, whereby the same variables were retained in both the donor and recipient datasets, the matching and non-matching variables were directly comparable. This is illustrated in Supplementary Fig. 1. In typical statistical matching studies, the donor dataset A contains variables X and y, and the recipient dataset B contains variables X and z, with X being the common variable used for matching. Typically, the donor and recipient datasets share only the matching variables (X) and not the target variables (y and z). However, in our simulation study, both the donor and recipient datasets contained the same set of variables, which enabled us to compare the distributions of matching and non-matching variables. Moreover, standardized mean differences were calculated to quantify the similarity between variables in terms of mean differences.

Evaluation metrics

Cox regression analysis was used to estimate the hazard ratios (HRs) for the time to incident diabetes. Univariate models with the main predictor and multivariate models with the main predictor and potential covariates were fitted. Potential covariates included age, sex, smoking status, hypertension, hemoglobin, cholesterol, and HbA1c levels. Additionally, BMI was included in the HOMA-IR/B model, and glucose was included in the BMI/WC

model. The covariate values were sourced from donor datasets. As a specific measure of data utility, we computed a confidence interval (CI) overlap measure to assess concordance²³. We calculated the overlap between the CIs for the matched data [L_M, U_M] and real data [L_R, U_R] using Eq. (1):

$$\frac{1}{2} \times \left(\frac{O_U - O_L}{U_R - L_R} + \frac{O_U - O_L}{U_M - L_M} \right) \times 100 \tag{1}$$

where [O_L, O_U] represents the overlapping interval. The overlap measures ranged from zero (no overlap) to one (perfect overlap). From the four categories of either the HOMA-IR/B or BMI/WC predictors, three different HRs and corresponding utilities were calculated. We present the AU from the three overlap measures. This computation was performed using the univariate Cox regression model. For prediction performance, Harrel's C-index was calculated using the multivariate Cox regression model. To account for the potential influence of multicollinearity on confidence interval overlaps, we calculated generalized variance inflation factor (gVIF)²⁴.

Results

Generation of synthetic data

From a total of 10,030 subjects, we included 8790 subjects (progressor-to-diabetes group [895 subjects, 10%] and nonprogressor group [7895 subjects, 90%]) with no missing baseline values. Synthetic data generation was applied to the entire dataset and a randomly selected quarter of the data, resulting in the following ten synthetic datasets: $S^{1_{100\%}}$, $S^{10_{100\%}}$, $S^{1_{25\%}}$, and $S^{10_{25\%}}$ (Fig. 1). The standardized mean differences across the covariates between the real and synthetic data were less than 0.03, indicating that the real and synthetic data exhibited similar univariate means (Table 1, Supplementary Table 1, and Supplementary Figs. 1 and 2).

Statistical matching

In the real data, high HOMA-IR/low HOMA-B and elevated BMI and WC were strongly associated with incident diabetes. The estimated HR in the univariate regression was 9.9 (95% confidence interval [CI]: 6.08, 16.1) for the HOMA-IR/B model and 3.45 (2.54, 4.70) for the BMI/WC model (Supplementary Table 2).

In each of the five matching schemes, we used nine sets of matching variables. As we evaluated how data utility was affected by the selection of matching variable sets and donor-recipient datasets, we observed that selecting a key matching variable was crucial.

Figure 2 illustrates how the cholesterol variables of matched individuals in the donor and recipient datasets are differently distributed based on the choice of matching variables. Statistical matching aims to balance cholesterol

	R _{25%}			S ¹ _{25%}			SMD
	DM (n = 212)	Non-DM (n = 1986)	p	DM (n = 223)	Non-DM (n = 1975)	p	
Age	55.4 (8.4)	52.0 (9.0)	< 0.01	54.7 (8.6)	52.1 (8.9)	< 0.01	< 0.01
Sex [Female]	116 (54.7)	1075 (54.1)	0.93	127 (57.0)	1058 (53.6)	0.37	0.01
Smoking [Yes]	94 (44.3)	759 (38.2)	0.1	87 (39.0)	758 (38.4)	0.91	0.01
Hypertension [Yes]	58 (27.4)	252 (12.7)	< 0.01	64 (28.7)	261 (13.2)	< 0.01	0.02
Glucose	93.6 (12.5)	82.9 (8.7)	< 0.01	90.1 (12.5)	82.9 (9.2)	< 0.01	0.02
Hemoglobin	13.7 (1.6)	13.5 (1.6)	0.02	13.7 (1.6)	13.5 (1.6)	0.17	0.01
Insulin	9.3 (7.7)	7.5 (4.1)	< 0.01	8.6 (8.1)	7.6 (4.7)	< 0.01	0.01
Cholesterol	197.8 (33.1)	191.0 (35.3)	0.01	197.2 (33.6)	190.0 (34.9)	< 0.01	0.03
HbA1c	6.1 (0.6)	5.5 (0.4)	< 0.01	6.1 (0.6)	5.5 (0.4)	< 0.01	< 0.01
BMI	26.0 (3.4)	24.5 (3.1)	< 0.01	25.4 (3.0)	24.5 (3.2)	< 0.01	< 0.01
HOMA-IR/B			< 0.01			< 0.01	0.04
Low/high	23 (10.8)	611 (30.8)		36 (16.1)	577 (29.2)		
Low/low	46 (21.7)	473 (23.8)		57 (25.6)	501 (25.4)		
High/high	89 (42.0)	776 (39.1)		86 (38.6)	771 (39.0)		
High/low	54 (25.5)	126 (6.3)		44 (19.7)	126 (6.4)		
BMI/WC			< 0.01			< 0.01	0.02
Low/low	68 (32.1)	1074 (54.1)		86 (38.6)	1072 (54.3)		
High/low	35 (16.5)	395 (19.9)		39 (17.5)	386 (19.5)		
Low/high	10 (4.7)	102 (5.1)		18 (8.1)	98 (5.0)		
High/high	99 (46.7)	415 (20.9)		80 (35.9)	419 (21.2)		

Table 1. Baseline characteristics of a quarter of the real data and the corresponding synthetic data. ¹A synthetic data. DM, diabetes mellitus; BMI, body mass index; HOMA-IR, homeostasis model assessment of insulin resistance; HOMA-B, homeostasis model assessment of beta-cell function; SMD, standardized mean difference; WC, waist circumference. Data are presented as mean (standard deviation) for continuous variables and n (%) for categorical variables. The p-values are derived from the t-test for continuous variables and the chi-squared test for categorical variables.

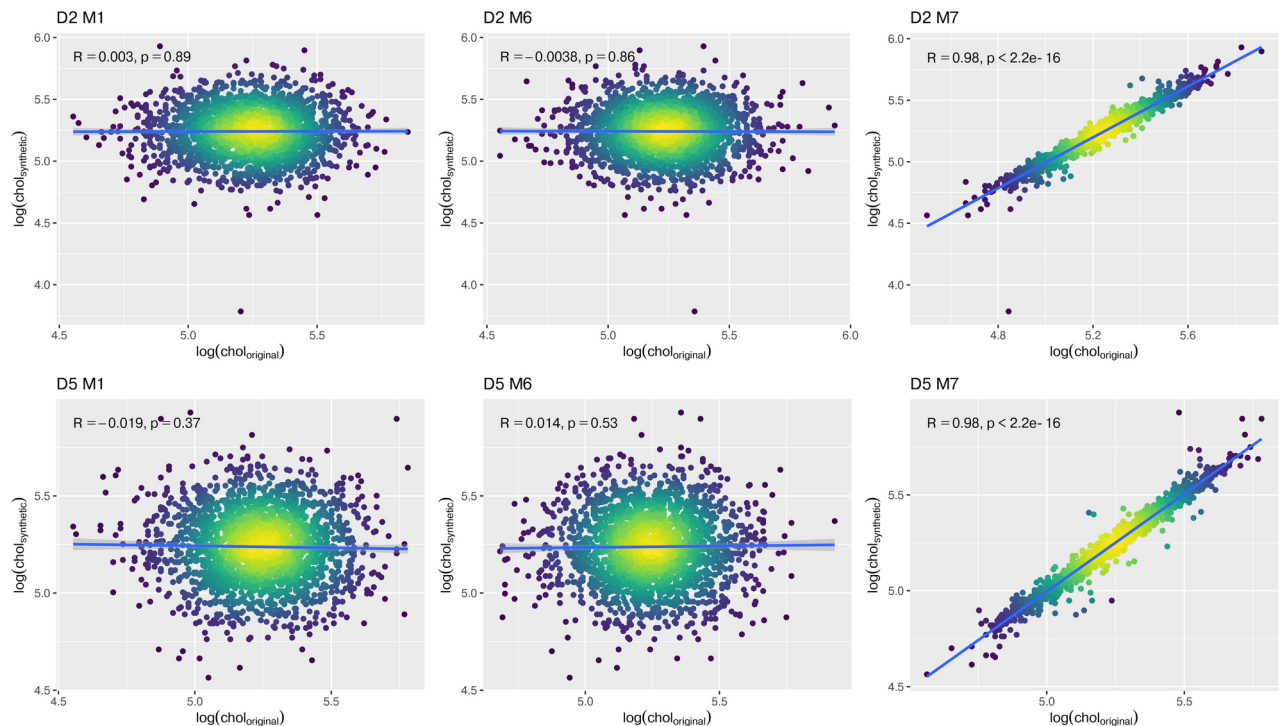


Fig. 2. Scatter plots of pairs of cholesterol variables from matched individuals in the donor and recipient datasets. The line represents perfect agreement ($y=x$). (A) The real-real donor-recipient scheme ($R_{75\%} \rightarrow R_{25\%}$); (B) The synthetic-synthetic donor-recipient scheme ($S_{100\%} \rightarrow S_{25\%}$). The left panel shows randomly selected pairs (low agreement), the middle panel shows pairs matched using only glucose (moderate agreement), and the right panel shows pairs matched with cholesterol, glucose, and BMI (high agreement). Higher Pearson's correlation R values suggest a stronger bivariate correlation.

values in both datasets while optimizing distances across multiple variables. Thus, our goal was to empirically show how closely cholesterol aligns under different matching scenarios. In the randomly selected pairs (M1) and pairs matched using only glucose (M6), the cholesterol variables for the donor and recipient datasets appeared to be different (left and middle panels). When cholesterol, glucose, and BMI (M7) were used for matching, the cholesterol variables of the matched individuals in the donor and recipient datasets had similar values (right panel). Notably, the synthetic-synthetic donor-recipient scheme ($S_{100\%} \rightarrow S_{25\%}$) showed almost identical results to those of the real-real donor-recipient scheme ($R_{75\%} \rightarrow R_{25\%}$) (upper panel versus lower panel). This strong association may weaken if a closely related but non-matching variable, such as WC, is included, as illustrated in Supplementary Fig. 3.

Data utility

The real-to-real pairs ($R_{100\%} \rightarrow R_{25\%}$) mimic record linkage with the identifier (D1 scheme in Supplementary Tables 1 and 4), whereby the same units are contained in both datasets. In this setting, including more variables in the matching process substantially increases data utility. For example, the real-to-real pairs ($R_{100\%} \rightarrow R_{25\%}$) in the HOMA-IR/B model (D1 scheme in Supplementary Table 3), including more variables in the matching process, result in similar HRs as those estimated in the real data (Supplementary Table 5). They also substantially increase data utility measured using CI overlap (median [interquartile range, IQR] average utilities [AUs] for matching using only glucose versus all available variables in the HOMA-IR/B model: 61% [61%, 61%] and 100% [100%, 100%] in $R_{100\%} \rightarrow R_{25\%}$ respectively). However, in donor-recipient pairs resembling statistical matching, adding more variables does not always lead to better performance. For example, in the $S_{100\%} \rightarrow S_{25\%}$ matching scheme, which represents the greatest departure from real data, the utility of data matched using only glucose is superior to that of the data matched using all variables except glucose (median [IQR] AUs for matching using only glucose versus all available variables except glucose for the HOMA-IR/B model: 70% [65%, 74%] and 21% [14%, 27%], respectively). In the BMI/WC model (Supplementary Tables 4 and 6), the real-to-real pairs show that an increased number of matching variables tends to enhance data utility (median [IQR] AUs in matching using height and weight versus all available variables in the BMI/WC model: 84% [84%, 84%] and 100% [100%, 100%] in $R_{100\%} \rightarrow R_{25\%}$, respectively). In statistical matching, data utility is generally higher when matched using all available variables (median [IQR] AUs for matching using height and weight versus all available variables in the BMI/WC model: 54% [53%, 66%] and 75% [60%, 76%] in $S_{100\%} \rightarrow S_{25\%}$, respectively). It is also observed that matching with more variables does not always result in superior performance (median [IQR] AUs for matching using height and weight as well as cholesterol, glucose, and BMI variables for BMI/WC model: 71% [64%, 76%] and 70% [62%, 77%] in $R_{100\%} \rightarrow S_{25\%}$, respectively, as shown in Supplementary Table 4).

In the multiple Cox regression analysis, models built using more matching variables showed improved prediction performance. Notably, synthetic data provided a median Harrell's C-index of 77% and 78%, lower than the 82% and 83% observed in real data from the HOMA-IR/B and BMI/WC models, respectively. While synthetic data could maintain reasonable prediction performance, further investigation is warranted to understand the performance difference (median [IQR] Harrell's C-index for matching using all available variables in the HOMA-IR/B model in $R_{100\%} \rightarrow S_{25\%}$, and $S_{100\%} \rightarrow S_{25\%}$: 79% [78%, 80%] and 77% [77%, 77%], as compared to 82% in real data; please see Supplementary Table 7. For the BMI/WC model in $R_{100\%} \rightarrow S_{25\%}$ and $S_{100\%} \rightarrow S_{25\%}$, these values were 80% [80%, 82%] and 78% [77%, 78%], as compared to 83% in real data; please see Supplementary Table 8). To address multicollinearity issues, we calculated the gVIF, and the results indicated that multicollinearity was not a notable issue.

Standardized mean differences

The discrepancy between the same set of variables from donor and recipient datasets was measured using standardized mean differences. A standardized mean difference of less than 0.01 was substituted with 0.01. Across covariates, matching schemes, and matching variable settings, the medians of the standardized mean difference were less than 0.03, indicating that the covariates of the donor and recipient datasets were similarly distributed. By matching the schema of the real data involving the donor and recipient datasets (M1 and M2), a small standardized mean difference was observed, with the exception of random pairs. Conversely, for the matching schema involving synthetic data (M3 to M5), the standardized mean differences increased. However, distribution discrepancy was suppressed when more matching variables were introduced (Supplementary Fig. 4).

Discussion

This study demonstrated how synthetic data combined with statistical matching can be used under plausible clinical conditions. First, we showed that data utility largely depends on the selection of matching variables in statistical matching. Although including more variables generally improves the explanatory power of the target variables, it is not always optimal. Increasing the number of matching variables can enhance the ability to explain target variables but also reduce the number of variables available for analysis. A potential matching variable (X) can capture the association between the target variables (y and z) to control for the effect of the covariate (y) on the primary variables of interest (z)²⁵. In our study, glucose, and HbA1c were greatly associated with the onset of diabetes (Spearman's rho coefficient for glucose and HbA1c were as follows: 0.071 and 0.101, respectively; see Supplementary Table 9). Accordingly, glucose or HbA1c could be considered suitable matching variables in either the HOMA-IR/B or BMI/WC setting. However, the results revealed that HbA1c was an ineffective matching variable for the HOMA-IR/B, while glucose was ineffective for the BMI/WC setting. Our findings are consistent with previous reports that highlight how matching variables are crucial for effective data utilization. When conducting statistical matching, it is advisable to select clinically relevant and statistically correlated variables. The quality of matching depends on identifying variables that satisfy the conditional independence assumption (CIA), which posits that matching variables (X) should account for the shared association between the target variables (Y and Z). Although CIA is often criticized for its strict selection criteria, prioritizing clinically significant variables and assessing their plausibility with available data can serve as a practical starting point. Additionally, conducting simulations, when sufficient data are available, can help empirically evaluate the effectiveness of different matching variable combinations, improving covariate balance and the utility of matched datasets^{2,20}.

Second, the synthetic data demonstrated acceptable data utility when compared to real data. Interestingly, models using synthetic data for both the recipient and donor datasets performed slightly better than those using synthetic data only for the donor dataset while matching all possible variables. This improvement may be attributed to the closely aligned multivariate distribution and inductive similarity of synthetic data, as they are generated through the same process. This alignment likely facilitates better matching between synthetic datasets when more variables are involved. However, we lack specific metrics to support this explanation. Third, we observed that generating high-quality synthetic datasets is a crucial step. We implemented nonparametric synthetic data generation using sequential classification and regression tree methods. Recent comparative analysis showed that tree-based synthesizers may offer a viable alternative to deep generative models for static tabular data^{26–28}. When the synthetic data was poorly constructed, the resulting estimates were also poor reflections of the actual data (data not shown). We only assessed univariate and bivariate similarities between original and synthetic data by evaluating standardized mean differences and correlation coefficients, respectively. To further evaluate multivariate distributional similarity, researchers can explore advanced evaluation methods such as propensity score mean-squared error, maximum mean discrepancy, the Mardia multivariate normality test, and covariance comparisons^{28–30}.

Utilizing individual data for biomedical research presents the ethical dilemma of privacy. Health records, which often contain a wide range of physical, clinical, or economic information, are highly likely to include private and personal details that can be linked to identifiable individuals. It is imperative for data providers and researchers to establish a balance between making data accessible and ensuring data privacy protection.

There is a growing body of evidence suggesting that synthetic data are useful in the biomedical field. For example, synthetic data generated by the UK Cohort National Cancer Registration and Analysis Service were used to test the effect of HPV on the reduction of cervical cancer in England³¹.

This research does not aim to prescribe methodological approaches for synthetic data generation and statistical matching, but rather to evaluate the empirical performance of these techniques under realistic conditions. While synthetic data have various uses, such as creating datasets for teaching, algorithm development, and testing, our study proposed its use for data integration from multiple sources. This application holds promise but also requires careful consideration of the limitations and potential risks of using synthetic data. As highlighted in

previous work, synthetic data should not replace real data for final analyses but rather serve as a useful tool during earlier stages of research.

Synthetic data generation methods were initially developed within the framework of statistical disclosure control and have since evolved with advancements in statistical modeling, probabilistic approaches, machine learning, and deep learning. Key approaches include multivariate normal distribution, Gaussian mixture models with Bayesian inference, tree-based synthesizers, and generative adversarial networks (GANs). Tree-based methods, such as CART, construct trees sequentially and draw samples, with extensions like random forests and XGBoost incorporating ensemble learning techniques. GANs, which involve two competing neural networks (a generator and a discriminator) have become central to synthetic data generation as deep learning techniques continue to advance. Variants of GANs, such as conditional GANs and deep convolution GANs, have been actively applied across various data types, including images, time-series data, and tabular data^{9,32}.

There are notable barriers to using synthetic data. First, studies have shown that synthetic data generated using methods such as CART does not fully eliminate disclosure risk, especially in cases where highly accurate models can inadvertently enable inference disclosure^{33,34}. Second, preserving missing data patterns in synthetic data remains challenging due to the unknown missingness mechanism and the complexity of estimating data distribution of both the underlying and missing data. Despite these challenges, some synthesizers, including the CART method used in this study, can generate synthetic data that reflect missing data, offering a potential solution when the missingness mechanism is partially understood^{35,36}.

Our study had several limitations. First, we utilized a single dataset, a specific synthesizer, and a specific model, which may limit the generalizability of our results. While a CART-based tabular data generation approach is well-established, deep learning-based synthetic data synthesizers are rapidly developing, and future research could further evaluate their potential to generate high-quality survival data. One promising approach for imputing time-to-event involves modeling the baseline hazard function using the given covariates³⁷. Second, we employed only one statistical matching method—the nearest neighbor hot-deck method—because it is popular in the statistical matching field^{4,38}. The exploration of other matching methods could provide further insights. Third, we included individuals with no missing data during the data synthesis process, which may have limited the generalizability of our findings. However, our results were similar to those obtained using synthetically generated data with missing values (data not shown).

Synthetic data generation offers a valuable alternative for facilitating large-scale biomedical research when access to original data is limited or restricted. Our study demonstrated that synthetic data can be effectively used for data integration, especially when using appropriate matching variables and methods. Statistical matching with synthetic data provides a promising solution for balancing privacy protection and data utility, thereby enabling meaningful research. Future research should include simulation studies to evaluate various synthetic data generation methods and synthesizer performances in complex data environments with diverse correlation structures.

Data availability

All data used in this study were obtained from the KOGES database, available at www.nih.go.kr. The credentialed users completed the required process and signed a data use agreement for the project.

Code availability

A code is available upon request from the corresponding author. All codes were written using R. The pseudocode can be provided as supplementary information.

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References

- Smith, D., Elliot, M. & Sakshaug, J. W. To link or synthesize? An approach to data quality comparison. *J. Data Inf. Qual.* **15**, 1–20 (2023).
- Gessendorfer, J., Beste, J., Drechsler, J. & Sakshaug, J. W. Statistical matching as a supplement to record Linkage: A valuable method to tackle nonconsent bias?. *J. Off. Stat.* **34**, 909–933 (2018).
- Drechsler, J. *Synthetic Datasets for Statistical Disclosure Control* (Springer, 2011).
- Kokosi, T. et al. An overview of synthetic administrative data for research. *Int. J. Popul. Data Sci.* **7**, 1727 (2022).
- Resnick, D. M., Cox, C. S. & Mirel, L. B. Using synthetic data to replace linkage derived elements: A case study. *Health Serv. Outcomes Res. Methodol.* **21**, 389–406 (2021).
- Dennett, A., Norman, P., Shelton, N. & Stuchbury, R. A synthetic longitudinal study dataset for England and Wales. *Data Brief* **9**, 85–89 (2016).
- Li, J., Cairns, B. J., Li, J. & Zhu, T. Generating synthetic mixed-type longitudinal electronic health records for artificial intelligent applications. *npj Digit. Med.* **6**, 98 (2023).
- Haendel, M. A. et al. The national COVID cohort collaborative (N3C): rationale, design, infrastructure, and deployment. *J. Am. Med. Inform. Assoc.* **28**, 427–443 (2021).
- Hernandez, M., Epelde, G., Alberdi, A., Cilla, R. & Rankin, D. Synthetic data generation for tabular health records: A systematic review. *Neurocomputing* **493**, 28–45 (2022).
- Tucker, A., Wang, Z., Rotalinti, Y. & Myles, P. Generating high-fidelity synthetic patient data for assessing machine learning healthcare software. *npj Digit. Med.* **3**, 147 (2020).
- Jo, H. et al. Insulin resistance and impaired insulin secretion predict incident diabetes: A statistical matching application to the two Korean nationwide, population-representative cohorts. *Endocrinol. Metab.* **39**, 711–721 (2024).
- Kim, Y., Han, B. G., KoGES group. Cohort profile: The Korean genome and epidemiology study (KoGES) consortium. *Int. J. Epidemiol.* **46**, e20 (2017).
- Song, Y. et al. Insulin sensitivity and insulin secretion determined by homeostasis model assessment and risk of diabetes in a multiethnic cohort of women: The women's health initiative observational study. *Diabetes Care* **30**, 1747–1752 (2007).

14. Haam, J. H. et al. Diagnosis of obesity: 2022 update of clinical practice guidelines for obesity by the Korean Society for the study of obesity. *J. Obes. Metab. Syndr.* **32**, 121–129 (2023).
15. Tabák, A. G. et al. Trajectories of glycaemia, insulin sensitivity, and insulin secretion before diagnosis of type 2 diabetes: An analysis from the Whitehall II study. *Lancet* **373**, 2215–2221 (2009).
16. Ohn, J. H. et al. 10-year trajectory of beta-cell function and insulin sensitivity in the development of type 2 diabetes: A community-based prospective cohort study. *Lancet Diabetes Endocrinol.* **4**, 27–34 (2016).
17. Yu, H. J. et al. Association of weight status and the risks of diabetes in adults: A systematic review and meta-analysis of prospective cohort studies. *Int. J. Obes.* **46**, 1101–1113 (2022).
18. Després, J. P. & Lemieux, I. Abdominal obesity and metabolic syndrome. *Nature* **444**, 881–887 (2006).
19. Nowok, B., Raab, G. M. & Dibben, C. Synthpop bespoke creation of synthetic data in R. *J. Stat. Softw.* <https://doi.org/10.18637/jss.v074.i11> (2016).
20. Moretti, A. & Shlomo, N. Improving statistical matching when auxiliary information is available. *J. Surv. Stat. Methodol.* **11**, 619–642 (2023).
21. D’Orazio, M., Di Zio, M. & Scanu, M. *Statistical matching* (Wiley, 2006).
22. D’Orazio, M. StatMatch: Statistical matching or data fusion. R package version 1.4.1. (2022).
23. Karr, A. F., Kohnen, C. N., Oganian, A., Reiter, J. P. & Sanil, A. P. A framework for evaluating the utility of data altered to protect confidentiality. *Am. Stat.* **60**, 224–232 (2006).
24. Fox, J. & Monette, G. Generalized collinearity diagnostics. *J. Am. Stat. Assoc.* **87**, 178–183 (1992).
25. Leulescu, A. & Agafitei, M. Statistical matching: A model based approach for data integration. 10–12 (Eurostat, 2013).
26. Fössl, E. & Drechsler, J. An Evaluation of Synthetic Data Generators Implemented in the Python Library Synthcity. In *Privacy in Statistical Databases* (eds Domingo-Ferrer, J. & Önen, M.) 178–193 (Springer Nature Switzerland, 2024).
27. Thees, O., Novák, J. & Templ, M. Evaluation of Synthetic Data Generators on Complex Tabular Data. In *Privacy in Statistical Databases* (eds Domingo-Ferrer, J. & Önen, M.) 194–209 (Springer Nature, 2024).
28. Livieris, I. E., Alimpertis, N., Domalis, G. & Tsakalidis, D. An Evaluation Framework for Synthetic Data Generation Models. In *Artificial Intelligence Applications and Innovations* (eds Maglogiannis, I. et al.) 320–335 (Springer Nature Switzerland, 2024).
29. Snoke, J., Raab, G. M., Nowok, B., Dibben, C. & Slavkovic, A. General and specific utility measures for synthetic data. *J. R. Stat. Soc. Ser. A* **181**, 663–688 (2018).
30. Heine, J., Fowler, E. E. E., Berglund, A., Schell, M. J. & Eschrich, S. Techniques to produce and evaluate realistic multivariate synthetic data. *Sci. Rep.* **13**, 12266 (2023).
31. Falcato, M. et al. The effects of the national HPV vaccination programme in England, UK, on cervical cancer and grade 3 cervical intraepithelial neoplasia incidence: A register-based observational study. *Lancet* **398**, 2084–2092 (2021).
32. Jozdani, S., Chen, D., Pouliot, D. & Johnson, B. A. A review and meta-analysis of generative adversarial networks and their applications in remote sensing. *Int. J. Appl. Earth Obs. Geoinf.* **108**, 102734 (2022).
33. Little, C., Elliot, M. & Allmendinger, R. Comparing the Utility and Disclosure Risk of Synthetic Data with Samples of Microdata. In *Privacy in Statistical Databases* (eds Domingo-Ferrer, J. & Laurent, M.) 234–249 (Springer International Publishing, 2022).
34. Hittmeir, M., Mayer, R. & Ekelhart, A. A baseline for attribute disclosure risk in synthetic data. In *Proceedings of the Tenth ACM Conference on Data and Application Security and Privacy*. 133–143.
35. Templ, M., Meindl, B., Kowarik, A. & Dupriez, O. Simulation of synthetic complex data: The R package simPop. *J. Stat. Softw.* <https://doi.org/10.18637/jss.v079.i10> (2017).
36. Templ, M. *Visualization and Imputation of Missing Values: With applications in R* (Springer Nature, 2023).
37. Mosquera, L. et al. A method for generating synthetic longitudinal health data. *BMC Med. Res. Methodol.* **23**, 67 (2023).
38. D’Albeto, R. & Raggi, M. Integrating rather than collecting: Statistical matching in the data flood era. *Stat. Pap.* **65**, 2135–2163 (2023).

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Author contributions

J.H.O., M.-J.P., H.K., C.M.S., and S.A. conceived and designed the study; E.J. managed the project, with supervisory assistance from J.H.O.; C.M.S., and S.A.; H.J. assisted the project implementation and data collection; E.J. and S.A. performed the analyses; S.A. drafted the manuscript; J.H.O., M.-J.P., H.K., C.M.S., and S.A. reviewed and revised the manuscript for intellectual content. All authors read and reviewed the results, approving the final manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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